

Report

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Instructions to run

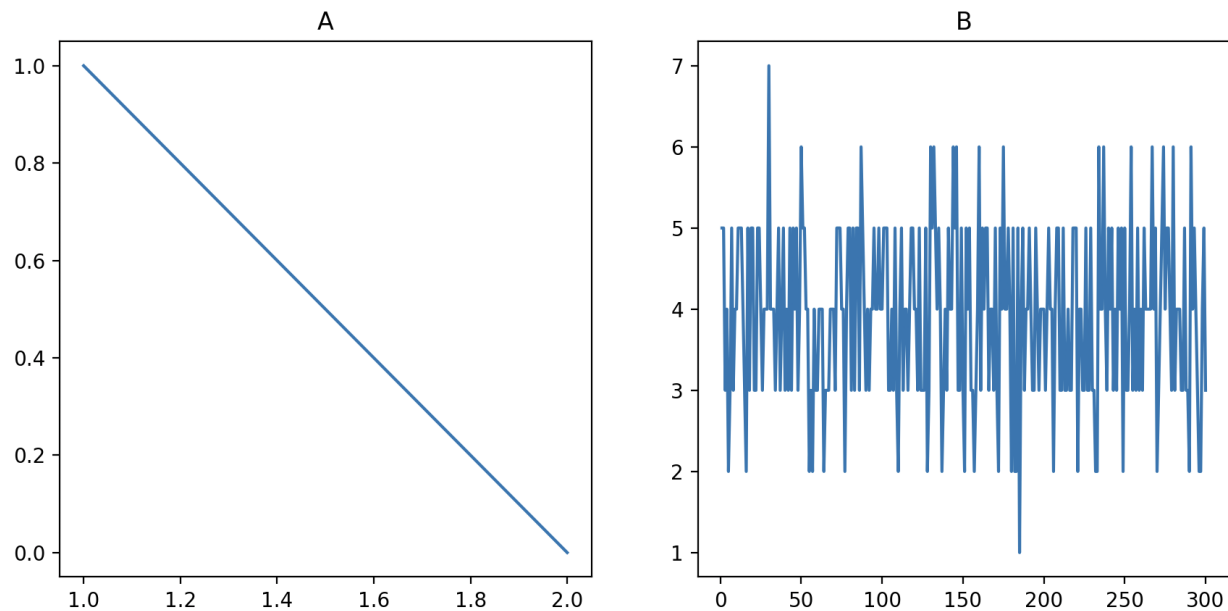
Please install these libraries depending on your system install commands

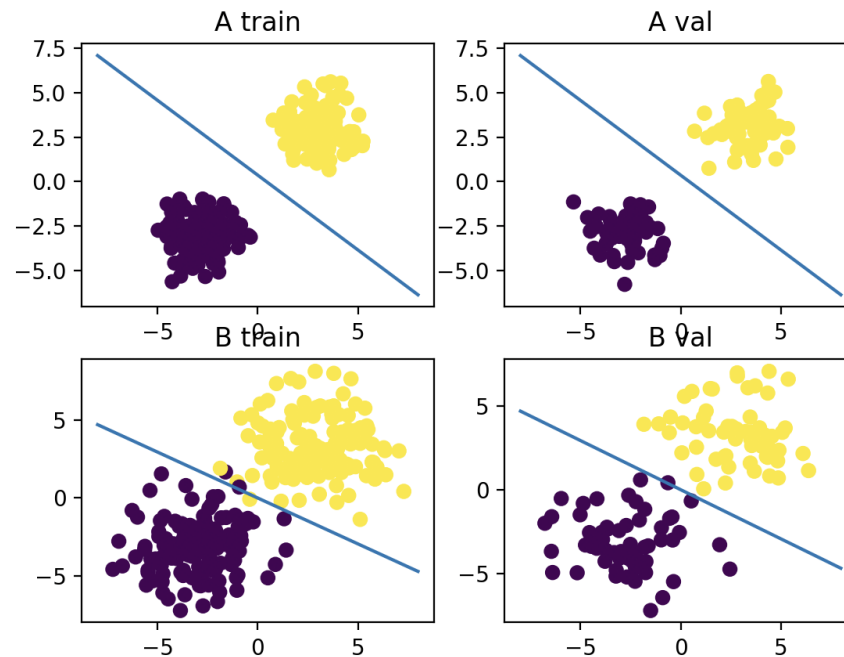
numpy matplotlib

I have written all the code in the .py file, so simply run it. I have also shown the plots in the .py file.

Q3 Interpretation and Output

The perceptron early stops for A since the datapoints are clearly linearly separable, so it detects a hyperplane and doesn't require all the epochs. But for dataset B which has the same mean but higher covariance, the datapoints become more intermixed and a clear hyperplane is not possible (or if it is possible requires more epoch than).





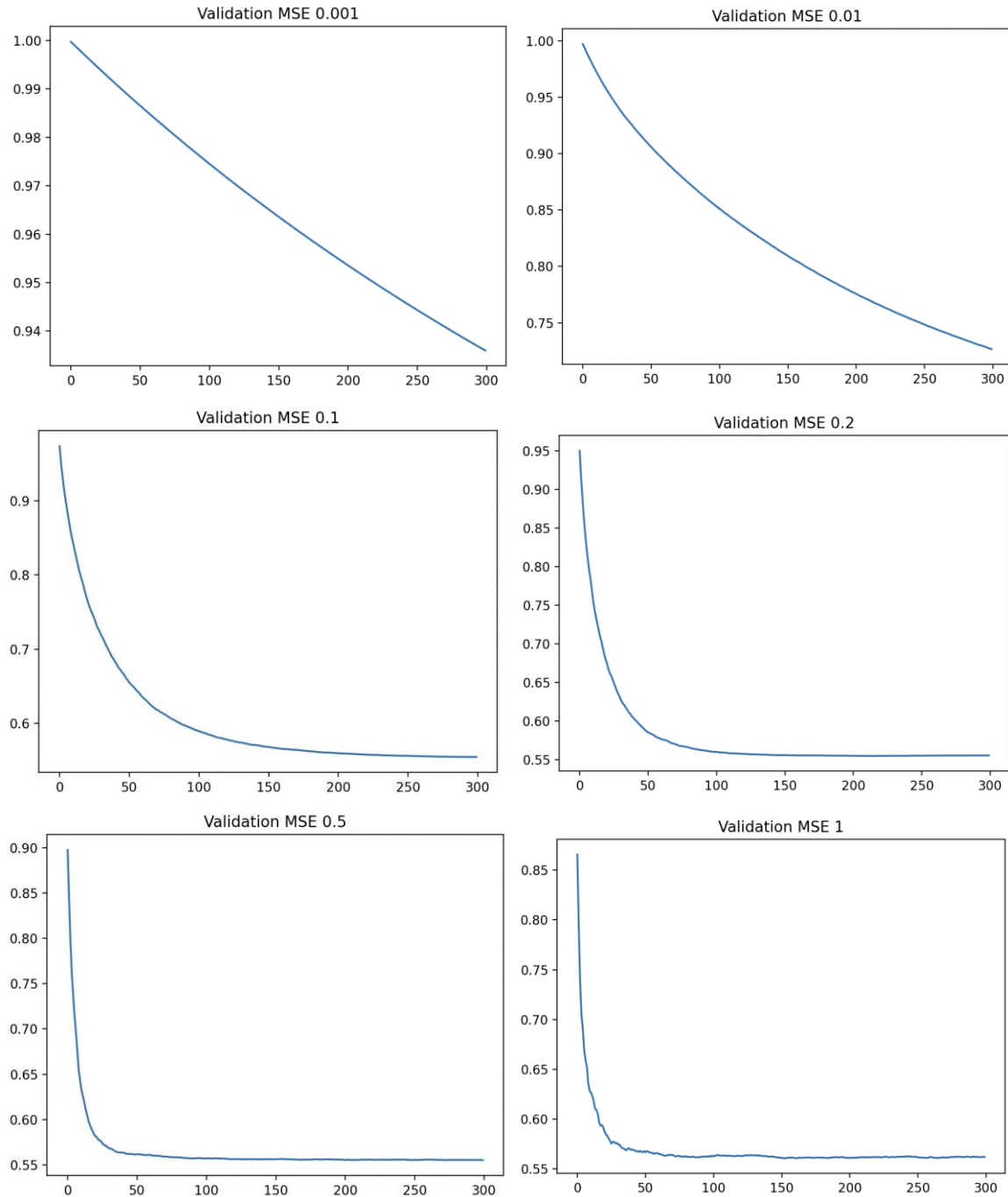
Early stop for A at epoch 2

Test accuracy for A: 1.0

Test accuracy for B: 0.9916666666666667

Q2 Interpretation and Output

For low values of n the curve doesn't steeply decrease, it is decreasing slowly. For larger values the curve is more aggressive and reaches the minimum earlier. There is small gradual decrease in the MSE. But for large values of n training a large number of trees is not suitable since it doesn't reduce the MSE which is because of overfitting. The curve is also more jagged.



n is 0.001

The minimum val mse iteration is 300

with mse 0.9359861264724341

Test mse on this is 0.9334820703173768

Train mse on this is 0.9337792218515711

n is 0.01

The minimum val mse iteration is 300

with mse 0.7265242107804213

Test mse on this is 0.7178195644807412

Train mse on this is 0.7107496841347711

n is 0.1

The minimum val mse iteration is 298

with mse 0.5545305667150862

Test mse on this is 0.5480113701820449

Train mse on this is 0.5403278542126102

n is 0.2

The minimum val mse iteration is 215

with mse 0.554560153216764

Test mse on this is 0.5467971843084749

Train mse on this is 0.538765306717881

n is 0.5

The minimum val mse iteration is 273

with mse 0.5550546765306367

Test mse on this is 0.548239122216805

Train mse on this is 0.5373370686564717

n is 1

The minimum val mse iteration is 153

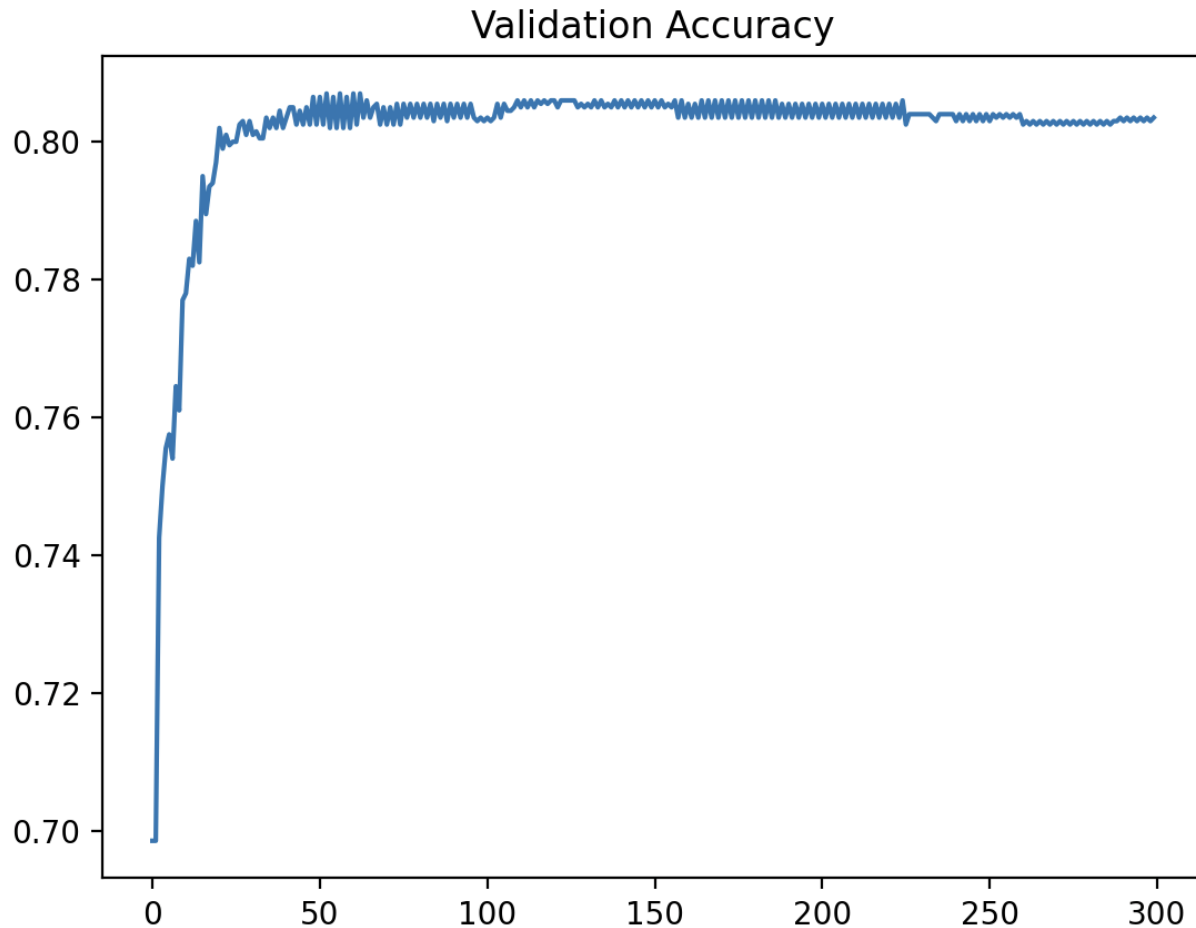
with mse 0.5607248069561922

Test mse on this is 0.5429459370176357

Train mse on this is 0.5377315365790617

Q1 Interpretation and Output

So as the number of trees grow we can see that the validation accuracy increases but reaches an upper bound. This is happening because of overfitting. Adaboosting is working fine after some trees to give a good accuracy.



The maximum val accuracy iteration is 53

with accuracy 0.807

Test accuracy on this is 0.8086388749372175